

From Boundary Crisis to Existential Law

A Worked Example of Boundary-Driven Abductive Locking

—*With a Concurrent Discussion of the Discovery of the Interactional Co-constitution
Theorem of Cognition and Existence*

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A Note on Genre

This paper is written in a genre that consciously revives the classical philosophical style—exemplified by Descartes' *Meditations on First Philosophy* or Hume's *An Enquiry Concerning Human Understanding*—in which the driving force is a genuine puzzlement rather than a gap in the scholarly literature, and in which every step of reasoning is openly submitted to the reader's scrutiny. It deliberately refuses unanchored jargon. Every technical term used in this paper is explained in ordinary language upon its first appearance and collected in the Conceptual Preliminaries that follow this note.

This paper does not request to be judged in the court of old academic conventions. It invites the reader to serve as an independent rational arbiter, reaching their own judgment based solely on the coherence and explanatory power of the argument itself.

How to Read This Paper

The core argument of this paper is a complete operational demonstration of Boundary-Driven Abductive Locking. Readers may follow these steps to scrutinize it:

1. What problem does this paper aim to solve? (Chapter 1)
2. What answers have predecessors given? What truths did they grasp, and what could they not explain? (Chapter 2)
3. What method does this paper employ to seek a new answer? (Chapter 3)
4. How does this method operate concretely in this case? (Chapter 4)
5. What is the final answer? How is it superior to previous answers? (Chapter 5)
6. Where does this answer point? (Chapter 6)
7. Does this paper's own writing process obey the law it discovers? (Chapter 7)

If the reader discovers a flaw at any step—an omission in the possibility space, an unfair criterion, a loose exclusion, a failed pressure test—then this paper's conclusion should be revised or abandoned. This is precisely the spirit of Boundary-Driven Abductive Locking.

Abstract

The debate between realism and constructivism concerning the relation between cognition and existence has persisted for over two millennia without resolution. This paper diagnoses the root of this impasse as an unexamined presupposition shared by both parties: the subject-object bifurcation. To break through the impasse, the question must be reframed from the more fundamental perspective of inter-systemic interaction.

This paper employs "Boundary-Driven Abductive Locking" as its methodological tool and presents a complete operational demonstration of the discovery of a meta-theoretical theorem. The operational steps include: (1) identifying the shared structure underlying the impasse; (2) executing a logical reversal—from "How does the subject know the object?" to "How do cognition and existence co-present in interaction?"; (3) delineating four logically possible positions (realism, constructivism, transcendental philosophy, and interactional co-constitution) and excluding them one by one according to three criteria; (4) confirming interactional co-constitution as the sole coherent position through pressure tests including the stone thought experiment, interaction-type analysis, and quantum entanglement boundary examination; (5) formalizing the confirmed position as the "Interactional Co-constitution Theorem of Cognition and Existence" (Meta-Theorem II of System Ontology).

This theorem asserts: existence and cognition are not independent entities, but rather two inseparable facets of inter-systemic interaction. What we call the "attributes" of existence are patterns stably presented in interaction; what we call "cognition" is the state-change induced within a particular system by interaction. The subject-object bifurcation is a cognitive illusion produced by the operation of symbolic encapsulation.

This paper makes a threefold contribution: (1) it discovers and formalizes a meta-theorem that resolves the millennia-old realism-constructivism debate; (2) it provides a complete operational demonstration of Boundary-Driven Abductive Locking, proving it to be a publicly verifiable tool for meta-theoretical discovery; (3) it furnishes a new ontological foundation for philosophy, cognitive science, and civilizational diagnostics (including such cognitive pathologies as nihilism and conceptual suspension). The writing of this paper itself constitutes a praxis of the law it discovers—refusing suspended symbolic manipulation and anchoring itself in the genuine interaction between cognition and existence.

Keywords: Boundary-Driven Abductive Locking; cognition and existence; interactional co-constitution; meta-theorem; subject-object bifurcation; system ontology

Conceptual Preliminaries

Before entering the main body of this paper, I will provide a concentrated exposition of the core terms that will appear repeatedly and the basic philosophical commitments that underpin the argument. This is not "academic jargon" but a transparent display of the cognitive tools employed in what follows. The reader may treat this section as a map, returning to it whenever encountering these terms in the main text.

I. Core Terms

1. Boundary-Driven Abductive Locking (Abductive Locking for short)

A method for discovering meta-theoretical theorems that proceeds from boundary crises, through logical reversal and the stepwise exclusion of a possibility space. It is not deduction (deriving conclusions from axioms), nor induction (deriving regularities from cases), but *abduction*—inferring the optimal explanation backward from phenomena. "Locking" refers to the procedure of excluding all logically possible positions one by one until only a single coherent position remains.

Everyday Anchoring: A forensic investigator inferring the characteristics of a criminal from traces left at the scene employs abductive reasoning. This paper's method infers the necessary conditions for a system's continued existence from the pattern of its breakdown at the limit—and upgrades this everyday reasoning into a rigorous procedure for philosophical discovery.

2. Boundary Crisis

When a system—whether a mathematical formal system, a physical theory, a social institution, or a cognitive framework—is pushed toward its operational limit (pursuing absolute completeness, unlimited subdivision, or total control), it triggers collapse or paradox. Boundary crises constitute the empirical starting point of this method.

Everyday Anchoring: Like a rubber band that snaps when stretched to its limit—the manner of snapping reveals, in reverse, the material and structure of the band. A boundary crisis is the "manner of snapping" of a system at its limit, and it reveals, in reverse, the necessary conditions for the system's continued existence.

3. Possibility Space

For a specific question, the set of all logically conceivable positions. A crucial step in Abductive Locking is first to exhaust this space, ensuring that no logically possible answer has been omitted.

Everyday Anchoring: In solving a case, one first lists all persons with possible motive and opportunity. If the true culprit is not on the list, all subsequent reasoning is in vain.

4. Subject-Object Bifurcation

The default presupposition that treats "the knower" (subject) and "the known" (object) as two independent entities. This paper argues that this very presupposition is the root of the problem, not its starting point.

Everyday Anchoring: We naturally feel that "I" and "the stone" are two fundamentally different kinds of thing. This paper asks: is this "natural feeling" itself an illusion that requires explanation?

5. Interactional Co-constitution

The core claim of the meta-theorem discovered and argued for in this paper: existence and cognition are not two independent entities, but two inseparable facets of inter-systemic interaction. The "hardness" of a stone is a pattern stably presented in the interaction between the stone and systems such as fingers or hammers; human cognition of "hardness" is the state-change induced within the human system by these interactions. Both co-present within a single current of interaction.

Everyday Anchoring: Consider the event of a handshake. The "firmness of the grip" is neither an attribute of your hand nor an attribute of the other person's hand; it is a pattern co-presented when the two hands interact. The relation between cognition and existence is, at bottom, a handshake.

6. Symbolic Encapsulation

The operation by which humans compress, name, and transmit stable patterns of interaction using symbols—words, formulas, diagrams. Symbolic encapsulation is a magnificent tool of human cognition, but it has a side effect: once encapsulation is complete, we tend to operate directly on the symbols, forgetting the concrete, bodily interactions that lie behind them.

Everyday Anchoring: Like the contact names in a mobile phone. The name "Mom" encapsulates the entire complex history of interactions between you and that specific person. The name is a tool, but if you merely shout at the name without dialing the number, you have mistaken the symbol for the thing itself.

7. Meta-Theorem

Within the system of System Ontology, a meta-theorem is an integrative statement situated between the meta-axioms (the deepest presuppositions) and the system axioms (the constitutive laws of concrete systems). It reveals global dynamical patterns generated by the coupled operation of the meta-axioms and system axioms. The "Interactional Co-constitution Theorem of Cognition and Existence" discovered in this paper is the second meta-theorem of System Ontology.

Everyday Anchoring: If the meta-axioms are the "basic rules" of board games (taking turns, win/loss conditions), and the system axioms are the "specific rules" of particular games (capturing in Go, the blocking of the horse in Chinese chess), then a meta-theorem is a "dynamical feature common to all board games" (such as the structure of opening, midgame, and endgame). It does not prescribe specific moves, but describes a deep pattern shared by all games of the kind.

II. Basic Philosophical Commitments of This Paper

1. System-Centrism, Not Anthropocentrism

This paper does not proceed from the anthropocentric perspective of "How do humans know the world?" but from the more fundamental perspective of "How do systems interact?" The human being is merely one particular kind of system; cognition is merely one particular form of interaction. The collision of stone against stone and the human grasping of a stone are isomorphic at the level of "inter-systemic interaction."

2. Methodological Transparency

This paper does not require the reader to accept any "intuition" or "insight" that is not publicly verifiable. Every step of reasoning—the delineation of the possibility space, the establishment of criteria, the exclusion of positions, the design of pressure tests—is open to public scrutiny. The reader may disagree with the conclusion, but if they wish to refute it, they must point to a specific flaw in the operations.

3. Conclusional Corrigibility

The meta-theorem discovered in this paper is not "absolute truth." If someone points out an omission in the possibility space, an unfair criterion, a loose exclusion, or a failed pressure test, the conclusion of this paper should be revised or abandoned. The spirit of Abductive Locking is not to find the ultimate answer once and for all, but to approach, continuously and rigorously, the answer that is most coherent under current cognitive conditions.

4. Writing as Praxis

The writing of this paper itself constitutes a praxis of the law it discovers. This paper refuses academic

jargon because Meta-Theorem II reveals that symbolic encapsulation can conceal interaction. It proceeds from a problem rather than from the literature because Meta-Theorem II reveals that cognition should anchor itself in direct interaction with the world, not in suspended symbolic systems. The form and content of this paper are unified.

Chapter 1: Introduction—A Millennia-Old Question

Stone is hard.

This is true in every known human culture. You pick up a stone; it resists your fingers. You strike one stone against another; they produce a sharp sound. You throw a stone into water; it sinks.

These phenomena are so stable that we never think they require explanation. Stone is hard—is this not self-evident?

But the problem lies precisely in this "self-evidence."

When we say "stone is hard," what exactly are we saying? Are we saying that the stone *itself* possesses an attribute called "hardness," which is there regardless of whether anyone perceives it? Or are we saying that "hardness" is a pattern of experience produced when humans interact with stones—a pattern we mark with the symbol "hard" in order to transmit and coordinate it within a group?

The former answer is called **realism**. It accords with our most naive and direct intuition. Of course the stone itself is hard. Would it become soft if we stopped looking at it? This intuition is so powerful that realism has been the default truth throughout most of human intellectual history.

The latter answer is called **constructivism**. It points out a difficulty that realism struggles to evade: if "hardness" is an attribute of the stone itself, then why do human understandings of stone differ so markedly across cultures and historical epochs? Primitive humans knew that stones were hard, but they did not know that stones are composed of silicate minerals, nor where stone falls on the Mohs scale of hardness, nor how many protons reside in the nuclei of its atoms. Were these later "discovered" attributes waiting in the stone all along to be found? Or were they gradually "constructed" as human cognitive tools evolved?

The constructivist intuition is equally powerful. We do indeed grasp the world through language, instruments, and theoretical frameworks. Change the symbolic system, change the measuring tools, and the world we see changes. In this sense, we can never "directly" see the stone itself; what we see is always the stone already mediated by our cognitive frameworks.

Thus emerges one of the most stubborn impasses in the history of philosophy:

The realist says: The stone itself possesses fixed attributes; cognition is the discovery of these attributes.

The constructivist says: Humans can never bypass cognitive frameworks to make direct contact with the stone itself; the "stone" we discuss is always already a stone constructed by our frameworks.

Each position rests on irrefutable intuitions; each has blind spots it cannot explain. Realism can explain why the hardness of stone is cross-culturally consistent (because the stone itself is hard), but struggles to explain why the scientific understanding of stone has undergone such marked historical evolution (if the attributes were there all along, why did it take us so long to discover some of them?). Constructivism can explain the historical variation of cognition (because cognitive frameworks evolve), but struggles to explain why certain cognitions are astonishingly consistent across all cultures (if everything is constructed, why has no culture ever constructed "stone is soft"?).

This impasse has persisted for over two thousand years. From Plato's theory of Forms to Aristotle's doctrine of substance, from Locke's "primary qualities" to Berkeley's "esse est percipi," from Kant's "thing-in-itself" to Saussure's "arbitrariness of the sign," from Wittgenstein's "language-games" to Foucault's "discursive construction"—thinkers have pulled back and forth on this question, yet neither side has ever succeeded in finally persuading the other.

This paper's diagnosis is this: **the reason this impasse cannot be resolved is not insufficiency of evidence or imprecision of argument, but rather that both parties share an unexamined**

presupposition—that subject and object are two independent entities, and the only question concerns their relation.

This presupposition is so deeply entrenched that we hardly ever question it. We naturally feel that "I" and "the stone" are two fundamentally different kinds of thing. I am the knower; the stone is the known. My task is to find the right way to know the stone.

But if—and this "if" is what this paper will argue—this presupposition is itself mistaken?

What if "subject" and "object" are not primordial forms of existence at all, but rather the products of our taking a more fundamental process—inter-systemic interaction—and retrospectively encapsulating it into two entities, only then to begin asking about their relation?

If this is so, then the millennia-old debate between realism and constructivism has been asking the wrong question from the very start. The question is not "How does the subject know the object?" but rather **"How do cognition and existence co-present in inter-systemic interaction?"** Under this new question, the truths that realism and constructivism each grasp can be unified within a more fundamental framework; their respective blind spots will dissolve as this framework is established.

This is what this paper undertakes to do.

This paper will employ a particular method of discovery—**Boundary-Driven Abductive Locking**—to accomplish this task. This method proceeds neither by deduction from given axioms nor by induction from numerous cases, but by identifying patterns in boundary crises, executing a logical reversal, and excluding a possibility space step by step, thereby inferring backward to the necessary conditions for a system's continued existence. This paper will take "the relation between cognition and existence" as a case study and present a complete operational demonstration of this method.

Specifically, this paper will proceed through the following steps:

First, we will analyze in finer detail the structure of the realism-constructivism impasse, identify the presupposition shared by both parties, and locate why this presupposition generates the impasse (Chapter 2).

Second, we will briefly state the operational steps of Abductive Locking, so that the reader understands the method by which this paper seeks a new answer (Chapter 3).

Third, we will strictly follow the steps of Abductive Locking to perform a complete discovery operation on "the relation between cognition and existence": delineate all logically possible positions, establish criteria for exclusion, exclude them one by one, apply pressure tests, until only a single coherent position remains (Chapter 4).

Fourth, we will formalize the remaining position as a meta-theorem—"The Interactional Co-constitution Theorem of Cognition and Existence"—and elaborate in detail its content, its cross-disciplinary anchorings, and its philosophical implications (Chapter 5).

Fifth, we will situate this meta-theorem within the history of philosophy, engaging in deep dialogue with ancestors such as Kant, Saussure, and Wittgenstein, clarifying what it inherits, what it revises, and where it transcends (Chapter 6).

Sixth, we will state the contributions and significance of this paper, and honestly indicate its boundaries and corrigibility (Chapter 7).

Finally, we will conduct a reflexive verification: does this paper's own writing process obey the meta-theorem it discovers? If the answer is no, then this paper is self-refuting; if the answer is yes, then the form and content of this paper constitute a unity (Chapter 8).

Before entering the main body, one matter requires advance clarification. This paper does not request to be judged in the court of old academic conventions—that is, it does not request the reader to assess its

"correctness" by appeal to any pre-existing philosophical position (whether realism, constructivism, or transcendental philosophy). For the argument of this paper points precisely toward the conclusion that each of these positions has its place within a more fundamental framework, but none of them suffices as the standard by which to judge that more fundamental framework.

What this paper invites the reader to do is this: **serve as an independent rational arbiter, follow each step of the reasoning, and scrutinize the soundness of each step.**

Has the possibility space been exhausted? Are the exclusion criteria fair? Is the exclusion operation rigorous? Do the pressure tests pass? If the reader discovers a flaw at any step, this paper's conclusion should be revised or abandoned.

This is not modest rhetoric; it is the inner spirit of Boundary-Driven Abductive Locking. It does not claim to have found "absolute truth"; it claims only that, given the delineated possibility space and criteria, the position that remains is the most coherent, most parsimonious, and most explanatorily powerful answer under current cognitive conditions. If someone can point to a better answer—whether because the possibility space was incomplete, the criteria require adjustment, or a pressure test was failed—then this paper will have fulfilled its task: it will have served as a publicly verifiable target that stimulated the birth of a superior theory.

Now, let us begin with that millennia-old question.

Chapter 2: The Structure of the Impasse—Truths and Limits of Existing Theories

To break through an impasse that has persisted for two millennia, one must first understand its structure with precision. The task of this chapter is not to list every view in the history of philosophy concerning the relation between cognition and existence, but rather to identify **what truths each party to the debate has grasped, and what each has failed to explain**, and thereby to locate **the unexamined presupposition they all share**.

2.1 Realism: The Hardness of Stone Belongs to the Stone Itself

Realism is humanity's most naive and direct intuition. Its core claim can be stated as follows:

Existence itself has a fixed structure. Cognition is the passive reflection or active discovery of this structure. The hardness of a stone belongs to the stone itself, regardless of whether anyone perceives it. The force of this intuition comes from everyday life. You pick up a stone; it resists. You put it down; it remains there. You leave the room and return; it is still the same stone. Nothing suggests that the stone's existence depends on your perception. On the contrary, your perception seems to be determined by the stone—as long as your senses are functioning normally, you "cannot help but" perceive it as hard.

Representatives of realism in the history of philosophy include Aristotle's doctrine of substance (things are composed of form and matter; form is the essential structure of the thing itself), Locke's doctrine of "primary qualities" (size, shape, motion are properties of the object itself, independent of the perceiver), and early Russell's logical atomism (the world is composed of logically independent "facts"; propositions are true or false depending on whether they correspond to these facts).

The truth realism grasps: cross-cultural consistency of cognition.

Why are stones hard in all cultures? Why does the law of non-contradiction hold in all cultures? Realism gives the most direct answer: because these attributes belong to the world itself. The world does not alter its structure because of cultural differences. Human cognition, so long as it is sufficiently accurate, will

necessarily converge upon these stable attributes. Realism provides the firmest intuitive foundation for the objectivity of knowledge.

What realism cannot explain: historical variation in cognition.

If all the attributes of the stone reside within the stone, waiting to be discovered, then why has the human scientific understanding of stone undergone such marked historical evolution? Primitive humans knew stone was hard, but did not know its chemical composition; Newton knew the mass of stone, but did not know mass-energy equivalence; we know far more than they did. If these later "discovered" attributes were in the stone all along, why is the order of their discovery not random? Why must certain discoveries await the maturation of particular technological conditions (microscopes, mass spectrometers, particle colliders)?

Realism may answer: "Because humans were not smart enough before, or their tools were not good enough." But this answer does not explain why "not good enough" tools correspond to "undiscoverable" attributes. The improvement of tools itself depends upon changes in the way humans interact with the world. If cognition is merely passive reflection, why would the content of the reflection change as the mode of interaction changes?

A more fundamental difficulty: realism cannot say what the "stone itself" stripped of all concrete modes of interaction actually is. Every time you perceive the stone, it is through some concrete interaction—touching with your hand, seeing with your eyes, measuring with an instrument. All the attributes you can describe are products of these interactions. If you strip away all these interactions, does the remaining "stone itself" possess any attributes at all? If not, what is it?

2.2 Constructivism: The Stone Is Carved Out by Symbolic Systems

Constructivism grows precisely in this blind spot of realism. Its core claim can be stated as follows:

Humans can never bypass cognitive frameworks to make direct contact with "existence itself." The "reality" we discuss is always already a reality constructed by symbolic systems—language, categories, social consensus. The hardness of a stone is not an attribute independent of symbolic systems, but the product of symbolic systems carving up and naming continuous experience.

The constructivist intuition is equally powerful. Consider: does a person who has never learned the word "hardness" have the same experience when facing a stone as someone who has? The former may feel a certain resistance, but he cannot cleanly separate this feeling from other feelings, nor can he use the concept "hardness" to compare the difference between this stone and another stone. Concepts are not neutral labels; they shape what we notice, what we ignore, how we categorize.

Representatives of constructivism in the history of philosophy include Saussure's structural linguistics (the meaning of a sign comes not from correspondence to things, but from relations of difference within the sign system), the later Wittgenstein's "language-games" (the meaning of a word lies in its use within concrete life-practices, not in its referent), and Foucault's theory of discourse (knowledge is not the discovery of objective truth, but the historical construction of power and discursive practices).

The truth constructivism grasps: historical variation and cultural diversity of cognition.

Why do humans in different eras and different cultures understand the world so differently? Constructivism gives a powerful answer: because their symbolic systems differ. The "Qi" of Chinese medicine and the "energy metabolism" of Western medicine are not different names for the same thing; they are two different ways in which two different symbolic systems carve up continuous experience. You cannot ask "What does 'Qi' correspond to in Western medicine?"—this question already presupposes lossless translation between two symbolic systems, whereas the core insight of constructivism is

precisely that different symbolic systems construct different "realities."

What constructivism cannot explain: cross-cultural consistency of cognition.

If everything is constructed by symbolic systems, why has no culture ever constructed "stone is soft"? Constructivists typically answer: "Because the physical structure of the stone limits the range of possible constructions." But this answer precisely concedes what constructivism needs to explain: this "physical structure"—is it not precisely that "existence itself" independent of symbolic systems? If existence itself exerts constraint upon construction, then construction is not arbitrary, and existence is not merely the product of construction.

A deeper difficulty: constructivism cannot explain why symbolic systems themselves evolve, or why the direction of evolution is not wholly random. If symbolic systems are merely arbitrary conventions, then the entire process from Aristotelian physics to Newtonian physics to Einsteinian physics should be nothing but a succession of conventions, with no "progress" to speak of. But this manifestly contradicts the self-understanding of scientific practice—scientists genuinely believe that relativity theory is "closer to the truth" than Newtonian mechanics, and not merely more convenient to use. Constructivism struggles to provide a convincing account of such "progress."

2.3 Transcendental Philosophy: Subject and Object Have a Common Root

Faced with the realism-constructivism impasse, the most ambitious attempt at reconciliation came from Kant. His solution can be summarized as follows:

The basic categories of cognition (time, space, causality) are neither derived from experience (contra realism) nor arbitrary conventions (contra constructivism), but are the a priori structures of the human cognitive subject. The world itself (the thing-in-itself) is unknowable, but humans organize experience through a priori categories, forming the "phenomenal world." Because all humans share the same a priori categories, scientific knowledge possesses universal necessity; because these categories are structures of the human subject, knowledge does not reflect the thing-in-itself.

Kant's solution marks a watershed in the history of philosophy. For the first time, he clearly demonstrated that cognition is not the passive reflection of an object by a subject, but the subject's active construction of experience. Yet he simultaneously preserved the universal necessity of knowledge—because the categories according to which construction proceeds are a priori structures shared by all humans.

The truth transcendental philosophy grasps: cognition is neither passive reflection nor arbitrary convention. It has structure, and this structure allows knowledge to be both universal and yet human.

What transcendental philosophy cannot explain: why do the a priori categories just happen to be compatible with the laws according to which the world operates?

This is the great pit Kant left behind. If the a priori categories are merely structures of the human subject, bearing no relation to the world itself (the thing-in-itself), then why does experience organized through these categories just happen to guide our successful action in the world? If causality is merely a category the human mind imposes upon experience, why do airplanes built using causal reasoning actually fly? Such "success" cannot be pure coincidence, yet Kant's theory cannot explain it.

Furthermore, Kant postulates a "thing-in-itself" that can never be tested by any experience. This postulate not only violates Ockham's razor (multiplying entities beyond necessity), but more importantly, it renders any discussion of "existence itself" forbidden territory. The subsequent German Idealism (Fichte, Schelling, Hegel) attempted precisely to eliminate this "thing-in-itself," but the path they took was to subsume subject and object under an even grander metaphysical entity (the Absolute Ego, Absolute Spirit). This path ultimately led to highly speculative systems that lost their anchoring in the empirical

world.

2.4 The Sprouts of Process Realism: Piaget and Merleau-Ponty

After Kant, there is a thread that comes closer to the final answer of this paper, yet it did not complete the breakthrough. The representatives of this thread are Piaget's genetic epistemology and Merleau-Ponty's phenomenology of the body.

Piaget demonstrated through extensive child psychology experiments that children's cognitive categories (space, time, causality, object permanence) are not a priori, but are gradually constructed through sensorimotor interaction with the environment. Through grasping, pushing, pulling, and throwing, the infant gradually differentiates subject and object out of an original "chaotic unity of self and world." In this sense, the subject-object bifurcation is not the starting point of cognition, but the **product** of cognitive development.

Piaget's insight was revolutionary: he historicized and geneticized Kant's "a priori categories." But he did not elevate this insight to the ontological level—he answered "How does the human individual acquire cognitive structures?" but did not answer "What is the relation between cognition and existence itself?"

Merleau-Ponty went further. He showed that perception is not the representation of an object by a subject, but a primordial, pre-reflective contact between the body and the world. In perception, subject and object are not yet differentiated. He used the concept of the "body-subject" to replace the Cartesian pure conscious subject, emphasizing that cognition is always embodied and situated.

Merleau-Ponty came very close to the conclusion this paper will ultimately reach. But he remained at the level of phenomenological description—he described the "what" of perceptual experience, but did not formalize it as a theorem about existence itself. He did not answer: if the subject-object bifurcation is derivative, then what is the underlying dynamics of that more primordial "body-world" intertwining?

2.5 The Root of the Impasse: The Unexamined Presupposition of the Subject-Object Bifurcation

We can now locate with precision the structure of this millennia-old impasse.

Realism grasps the constraint that existence exerts upon cognition, but cannot explain the historicity of cognition, because it conceives "existence" as a closed, static entity.

Constructivism grasps the activity and historicity of cognition, but cannot explain the constraint that existence exerts upon cognition, because it conceives "cognition" as a closed symbolic system.

Transcendental philosophy attempts reconciliation, but at the cost of introducing an untestable "thing-in-itself" or "Absolute Spirit," and cannot explain the compatibility between a priori structures and the laws of the world.

The sprouts of process realism have approached the answer, but have not completed the leap from phenomenological description to ontological theorem.

These four parties to the debate, despite their divergent positions, share a single unexamined presupposition: **subject and object are two independent entities, and the only question concerns their relation.**

Realism says: the object determines the subject.

Constructivism says: the subject constructs the object.

Transcendental philosophy says: subject and object have a common a priori root.

Process realism says: subject and object differentiate themselves out of a process.

But no one has asked: **are the very categories of "subject" and "object" themselves a cognitive**

illusion produced when a more fundamental process—inter-systemic interaction—is carved up by the operation of symbolic encapsulation?

If we demote "subject" and "object" from entities to facets of a process, then the original question—"How does the subject know the object?"—dissolves of itself. The new question is: **How do cognition and existence co-present in inter-systemic interaction?**

This is the logical reversal to be performed in the next chapter. But before performing this reversal, we must first introduce the tool we will use—Boundary-Driven Abductive Locking.

Chapter 3: Method—An Overview of Boundary-Driven Abductive Locking

Before entering the core argument, it is necessary to set forth the method this paper will employ. This is not to pile up methodological jargon, but to give the reader a clear sense of how the ensuing argument will unfold, what each operational step means, and at which points the reader may scrutinize the soundness of the reasoning.

The method employed in this paper is called **Boundary-Driven Abductive Locking** (hereafter, **Abductive Locking** for short). It is a systematic operational procedure for discovering meta-theoretical theorems that proceeds from boundary crises, through logical reversal and the stepwise exclusion of a possibility space. It is neither deduction from given axioms (the formal operation of mathematics), nor induction from numerous cases (the statistical operation of empirical science), but **abduction**—inferring the optimal explanation backward from phenomena.

3.1 Why Is a New Method Needed?

Traditional philosophical discovery has often relied upon the intuition of genius. Descartes, meditating beside his stove, hit upon "I think, therefore I am"; Kant, reasoning in his study, constructed a system of a priori categories. Such insights have profoundly altered human thought, but they share a common limitation: **the process of discovery is not publicly verifiable**. The reader is confronted only with the finished product—a completed philosophical system—and must then choose whether to accept it. As to how this system was derived step by step from the initial puzzlement, what other possibilities were excluded along the way, and why they were excluded—of this the reader remains ignorant.

This has had a consequence: philosophical disputes often devolve into clashes of position. Realists and constructivists each proceed from their own premises, derive their respective conclusions, and then accuse the other side of having faulty premises. Because neither side can demonstrate "why my premises are more reasonable than yours," the dispute can continue indefinitely.

Abductive Locking attempts to alter this situation. Its core promise is this: **to transform the process of meta-theoretical discovery from the private affair of genius into a publicly verifiable operational procedure**. It does not seek to eliminate the role of insight, but to compel insight to submit to the scrutiny of public reason at every step. Has the possibility space been exhausted? Are the exclusion criteria fair? Is the operation of exclusion rigorous? Do the pressure tests pass? Each of these questions the reader can judge for themselves, without needing to rely upon trust in the author's authority.

3.2 The Five Steps of Boundary-Driven Abductive Locking

The operation of Abductive Locking divides into five distinct steps. This section will provide an overview; Chapter 4 will use "the relation between cognition and existence" as a case study to demonstrate the concrete operation of each step.

Step One: Pattern Recognition in Boundary Crises

Any stable system, when pushed toward its operational limit—pursuing absolute completeness, unlimited subdivision, or total control—will trigger collapse or paradox. Gödel's incompleteness theorems in mathematics, the ultraviolet catastrophe in physics, Russell's paradox in set theory, and the paradox of control in social systems are all instances of boundary crises. A boundary crisis is not an accidental malfunction of a system, but the "developer fluid" for the conditions of a system's continued existence—at the very brink of collapse, it reveals in reverse the constraints that a system must satisfy in order to stably persist.

In the case of this paper, the boundary crisis is not mathematical or physical, but **the impasse of philosophical debate itself**. The realism-constructivism debate has persisted for over two millennia without resolution. This very "failure to resolve" is itself a boundary crisis—it suggests that the disputing parties may share a mistaken presupposition that locks the debate into an endless back-and-forth on a single plane.

Step Two: Logical Reversal

Traditional thinking asks: "Why does the system collapse at the boundary?" Abductive Locking executes a crucial reversal: **it asks instead, "Why can the system successfully persist most of the time?"**

This reversal transforms "stability" from a default background requiring no explanation into an achievement that positively demands explanation. If a system's continued existence is conditional, then what is extracted in reverse from boundary crises may well be those very conditions.

In the case of this paper, the traditional question is: "How does the subject know the object?" Abductive Locking reverses this into: "How do cognition and existence co-present in inter-systemic interaction?" This reversal demotes "subject" and "object" from entities to facets of a process, thereby dissolving the presupposition of the original question.

Step Three: Delineation and Locking of the Possibility Space

For the reversed question, one first **exhausts all logically possible answers**. This is the step requiring the greatest caution—if any possible position is omitted, the subsequent locking operation will rest upon that omission and be invalid.

Once the possibility space is exhausted, one **establishes criteria for exclusion**. The criteria must satisfy three conditions: they must derive from observable systemic behavior (not be purely speculative); they must apply fairly to all competing positions (not favor one side); and they must be as parsimonious as possible (Ockham's razor).

With the criteria established, one **excludes, one by one, the positions that fail to meet them**. The standard for exclusion is not "falsity" but "incoherence" or "explanatory redundancy"—that is, the position cannot explain crucial observational facts, or its core claim depends upon an additional presupposition that has not been tested and cannot be tested.

Step Four: Pressure Testing

The position that survives the locking procedure is then subjected to the most stringent boundary cases for testing. Pressure test cases typically include: limit situations of the theory itself, anomalous phenomena that appear to constitute counterexamples, and compatibility with other already-confirmed regularities.

If the surviving position passes the pressure tests, it is confirmed; if it fails, one must return to Step Three to check whether a possible position was omitted or whether the criteria require adjustment.

Step Five: Formalized Statement

The position that has passed the pressure tests is then formalized and stated as an axiom or theorem. The

requirements for formalization are: specify its dependencies (upon what more fundamental propositions it rests); its statement (the precise formulation of the core proposition); its explication (unfolding the meaning of the proposition); its cross-disciplinary anchorings (corresponding phenomena in other disciplines); and its philosophical dialogue (its relation to existing philosophical positions).

Formalization is not the concealment of the process of discovery, but the precise encapsulation of its result. It allows subsequent research to cite the theorem directly, without needing to repeat the entire discovery process each time.

3.3 The Epistemological Status of Boundary-Driven Abductive Locking

Abductive Locking is itself neither an axiom nor a theorem. It is **a method for discovering axioms and theorems.**

It does not claim to be the only method of discovery. But it does claim that the propositions obtained through this method are, given the delineated possibility space and criteria, the most coherent, most parsimonious, and most explanatorily powerful answers available within the current cognitive framework. Its openness lies in this: **if someone can point to an omission in the possibility space, an unfair criterion, a loose exclusion, or a failed pressure test, then the locking can be reopened and the conclusion revised.**

Boundary-Driven Abductive Locking is not a method for finding the ultimate truth once and for all, but a method for approaching, continuously and rigorously, the answer that is most coherent under current cognitive conditions.

Chapter 4: Core Argument—The Discovery of Meta-Theorem II

This chapter is the heart of the paper. I will strictly follow the five steps of Boundary-Driven Abductive Locking to present a complete demonstration of the discovery process of the "Interactional Co-constitution Theorem of Cognition and Existence." The reader may follow each step and scrutinize its soundness independently.

4.1 Pattern Recognition: The Deep Structure of the Impasse

In Chapter 2, we already identified the deep structure of the realism-constructivism impasse. Here I will offer a concise review, in order to establish this structure explicitly as the object of the first step of Abductive Locking.

Realism, constructivism, transcendental philosophy, and the sprouts of process realism—these four parties to the debate, despite their divergent positions, share a single unexamined presupposition: **subject and object are two independent entities, and the only question concerns their relation.**

- Realism says: the object determines the subject. Cognition is passive reflection.
- Constructivism says: the subject constructs the object. Cognition is active carving.
- Transcendental philosophy says: subject and object have a common a priori root, but the thing-in-itself is unknowable.
- Process realism says: subject and object differentiate themselves out of interaction, but does not inquire into the underlying dynamics of interaction.

None of the four parties has questioned whether the very categories of "subject" and "object" are primordial. They have taken for granted that "I" and "the stone" are two fundamentally different kinds of thing, and only then begun to debate "How do I know the stone?" This default presupposition is the root

of the impasse.

Conclusion of Pattern Recognition: To break through the impasse, one must suspend the presupposition of the "subject-object bifurcation" and return to a more primordial level—a level at which "subject" and "object" have not yet been differentiated.

4.2 Logical Reversal: From "Subject-Object Bifurcation" to "Interaction"

The traditional question: "How does the subject know the object?"

This very question already presupposes that subject and object are two already-differentiated entities. Whatever answer you give—reflection, construction, transcendental synthesis—you are already thinking within the acceptance of this presupposition. Any answer will therefore remain trapped in the back-and-forth between realism and constructivism.

Abductive Locking demands a crucial logical reversal. We cease to ask "How does the subject know the object?" and ask instead:

"How do cognition and existence co-present in inter-systemic interaction?"

The significance of this reversal is threefold:

1. **It demotes "entities" to "facets of a process."** The "subject" is no longer an independent entity, but the system that undergoes state-change within the interaction; the "object" is no longer an independent entity, but the pattern that stably presents itself within the interaction. The two are not two kinds of thing, but two inseparable facets of a single process of interaction.
2. **It dissolves the original question.** Once the "subject-object bifurcation" is revealed to be a derivative product of encapsulation, the original question—"How does the subject know the object?"—dissolves of itself. Just as, once "combustion" is revealed not to be the release of "phlogiston" but the combination of a substance with oxygen, the original question—"Where does phlogiston go during combustion?"—no longer requires an answer.
3. **It opens a new possibility space.** The new question allows us to ask: What structure does this "interaction" itself possess? What constraints does it obey? How does it operate at different levels? These questions will guide us toward deeper regularities.

Conclusion of Logical Reversal: This paper will no longer participate in the millennia-old back-and-forth between realism and constructivism. Instead, it will retreat to the step before their shared presupposition and begin anew from the more primordial level of inter-systemic interaction.

4.3 Delineation and Locking of the Possibility Space

4.3.1 Delineating All Logical Possibilities

For the reversed question—"How do cognition and existence co-present in inter-systemic interaction?"—we can conceive of four basic positions:

Position A (Realist-Type): Existence determines cognition. In interaction, the side of existence is active and determining; the side of cognition is passive and determined. The content of cognition is entirely fixed by the attributes of existence itself.

Position B (Constructivist-Type): Cognition constructs existence. In interaction, the side of cognition is active and carving; the side of existence is passive and carved. The "existence" we discuss is always already an existence constructed by cognitive frameworks.

Position C (Transcendental Philosophy-Type): Cognition and existence share a common a priori root. Interaction is possible because cognition and existence are homologous at some deeper level (the transcendental subject, Absolute Spirit). This common root guarantees the fit between cognition and

existence, but it cannot itself be tested by interaction.

Position D (Interactional Co-constitution-Type): Cognition and existence are two inseparable facets of interaction. There is no "existence prior to cognition," nor "cognition prior to existence." What we call the "attributes" of existence are patterns stably presented in interaction; what we call "cognition" is the state-change induced within a particular system by interaction. Both co-present within a single current of interaction.

Verification of the Possibility Space: Have these four positions exhausted the logical possibilities? We can test by elimination. Suppose there were a position that neither admitted that existence determines cognition (denying A), nor that cognition constructs existence (denying B), nor appealed to an untestable common root (denying C), nor admitted that cognition and existence co-constitute in interaction (denying D). Such a position could only claim that "cognition and existence have no relation whatsoever." But "no relation" would mean that cognition can never reach existence, that knowledge is utterly impossible. This position not only contradicts all human practice, but is self-refuting (as a knowledge claim, it must itself claim to have reached some existence). Therefore, the logically possible positions are exactly A, B, C, and D.

4.3.2 Establishing Criteria for Locking

We establish three criteria by which to filter the four positions:

Criterion One (Cross-Cultural Consistency of Cognition): A coherent position must be able to explain why certain cognitions are astonishingly consistent across all known human cultures. For example, why are stones hard in all cultures? Why does the law of non-contradiction hold in all cultures? If this consistency were merely "accidental consensus," its compelling force would remain unexplained.

Criterion Two (Historical Variation of Cognition): A coherent position must be able to explain why human cognition of the same kind of object varies significantly across different epochs. For example, why did primitive humans know that stone is hard, but not the atomic structure of stone? Why did Newton not know mass-energy equivalence? If cognition were merely passive reflection of a fixed existence, such historical variation should not exist.

Criterion Three (Ontological Parsimony): A coherent position should not postulate any additional entities that cannot be publicly tested. If two positions can explain the same phenomena, but one postulates an additional, untestable entity, the more parsimonious position should be chosen.

4.3.3 Exclusion by Criterion

Excluding Position A (Realist-Type):

Position A can pass Criterion One (the structure of existence itself guarantees cross-cultural consistency). But it cannot pass Criterion Two. If cognition were merely passive reflection of a fixed existence, human cognition of stone should be identical across all epochs—at most, there would be "more and more discoveries." But this "more and more" itself demands explanation: why is the order of discovery not random? Why must certain discoveries await the maturation of particular technological conditions? If the atomic structure of stone was in the stone all along, why could humans not "see" it before the invention of the electron microscope? Position A can only appeal to "tools were not good enough." But the essence of tools is the alteration of the way humans interact with the world. Position A cannot explain why a change in the mode of interaction would alter "the content reflected"—if the content was there all along, why would different modes of interaction "see" different contents?

Position A fails Criterion Two. **Excluded.**

Excluding Position B (Extreme Constructivist-Type):

Position B can pass Criterion Two (different symbolic systems construct different "realities," explaining historical variation). But it cannot pass Criterion One. If everything is constructed by symbolic systems, why has no culture ever constructed "stone is soft"? Constructivists typically answer: "Because the physical structure of the stone limits the range of possible constructions." But this answer already concedes the existence of a "physical structure" independent of symbolic systems that exerts constraint upon construction. If this constraint exists, then construction is not arbitrary, and existence is not merely the product of construction. Position B must either deny the compelling force of cross-cultural consistency (contrary to human experience) or surreptitiously introduce the very "existence itself" it criticizes in order to explain this force (self-contradiction).

Position B cannot pass Criterion One without self-contradiction. **Excluded.**

Excluding Position C (Transcendental Philosophy-Type):

Position C attempts to pass both Criterion One and Criterion Two simultaneously: the universality of a priori categories guarantees cross-cultural consistency; the fact that a priori categories are only structures of the subject (not reflecting the thing-in-itself) leaves room for historical variation. But Position C fails Criterion Three. It postulates a "thing-in-itself" (or "transcendental subject," or "Absolute Spirit") that can never be tested by any experience. More importantly, it cannot explain a crucial issue: if a priori categories are only structures of the human subject, unrelated to the thing-in-itself, then why does experience organized through these categories just happen to guide our successful action in the world? Why do airplanes built using causal reasoning actually fly? Such "success" cannot be pure coincidence. Position C must either admit some relation between a priori categories and the world (thereby collapsing back into A or D) or maintain the unknowability of the thing-in-itself while failing to explain the necessity of success.

Position C fails Criterion Three, and leaves an unresolved core difficulty within its own framework. **Excluded.**

Position D (Interactional Co-constitution-Type) Passes the Initial Screening:

Position D can pass Criterion One: cross-cultural consistency derives from the stability of interactional patterns. The interaction between stone and human fingers presents a pattern of resistance in all cultures; this pattern does not depend upon symbolic systems.

Position D can pass Criterion Two: historical variation of cognition derives from the evolution of modes of interaction. Primitive humans interacted with stone using their hands and obtained the pattern "hard"; modern humans interact with stone using electron microscopes and obtain the pattern "atomic structure." As the mode of interaction changes, the patterns stably presented change.

Position D can pass Criterion Three: it postulates no additional untestable entities. It describes only observable interactional processes and treats "cognition" and "existence" as two facets of this process.

Conclusion of Locking: A, B, and C are all excluded. D is the only position that passes the initial screening.

4.4 Pressure Testing

Position D, having passed the initial screening, must now be subjected to pressure tests at boundary cases. If it fails any test, we must return to the possibility space and re-examine whether any position was omitted or the criteria require adjustment.

4.4.1 Test One: The Stone Thought Experiment

Test Design: Imagine a hominid in a pre-linguistic era. He has no word for "stone," no concept of "hardness," no symbolic system whatsoever concerning stone. He sees a stone, picks it up, and feels resistance. He cracks open a fruit with the stone; the stone does not deform. This behavior contains a complete causal chain—from perception to decision to execution to outcome. The entire process depends upon no linguistic symbols.

Test Question: In this scenario, does the attribute "hard" of the stone exist?

Position D's Answer: It exists. But not as an independent attribute of the "stone itself," but as a pattern of the interaction between the stone and the hominid's fingers. The hominid does not need the symbol "hardness" to experience resistance in interaction. The symbol was invented later, to encapsulate and transmit this pattern. The pattern itself precedes the symbol.

Verification: A and C can also answer "it exists," but they cannot explain why the scientific understanding of "hardness" subsequently evolved. If B insists that "without symbols there are no attributes," it must answer: what was the resistance the pre-linguistic hominid experienced? If it was not the hardness of the stone, what was it? If B admits it was "pre-symbolic hardness," it has already moved toward D.

Test Conclusion: D passes. It provides the most parsimonious and coherent explanation: interactional patterns precede symbolic encapsulation.

4.4.2 Test Two: The Distinction Between Direct Interaction and Indirect Measurement

Test Design: Consider two scenarios.

Scenario One: A person directly picks up a stone with their hand and perceives its weight.

Scenario Two: A person uses a ruler to measure the length of the stone.

In Scenario One, the person interacts directly with the stone. In Scenario Two, the person interacts with the ruler, the ruler interacts with the stone, and the person infers an attribute of the stone by observing the state-change of the ruler.

Test Question: In Scenario Two, does it mean that the interaction between the person and the stone is "mediated"? Does this constitute a challenge to Position D—since the person seems not to interact directly with the stone, yet obtains cognition of the stone?

Position D's Answer: In Scenario Two, the person indeed has no direct physical contact with the stone. But the person interacts **indirectly** with the stone via the ruler. The ruler itself is a system, designed to interact with the stone in a specific type of interaction (length comparison) and to transform the result of this interaction into a state readable by the person (the scale). The act of reading the scale is the person's direct interaction with the ruler, by means of which the person indirectly obtains information about the ruler-stone interaction. The entire chain remains interaction; only the type of interaction is more complex.

A Deeper Explanation: Scenario Two is possible because the interactional pattern between ruler and stone and certain interactional patterns between person and stone share a single underlying rule—the measurement of spatial extension. This underlying rule is not constructed by any symbolic system; it is a pattern stably presented in countless inter-systemic interactions.

Verification: Position D not only explains Scenario Two, but also explains why "measurement" in Scenario Two reinforces the human illusion of subject-object bifurcation—because the person's attention is focused on the scale of the ruler (the intermediary system B) rather than on the direct experience of interaction with the stone. This "spectator perspective" makes it easier for the person to conceive of themselves as "subject" and the stone as "object." A and C cannot explain the origin of this illusion; B cannot explain why the interaction between ruler and stone is stable and repeatable.

Test Conclusion: D passes. It unifies the explanation of direct interaction and indirect measurement, and can explain why indirect measurement reinforces the illusion of subject-object bifurcation.

4.4.3 Test Three: Does Quantum Entanglement Constitute a Counterexample?

Test Design: In quantum mechanics, when two particles A and B are entangled, a measurement on A instantaneously affects the state of B, regardless of the distance between them. Does this imply a mode of existence "not dependent upon interaction"? If the state-change of B involves no interaction traveling from A to B (since information cannot be transmitted faster than light, yet the influence is instantaneous), is the foundation of Position D—"to exist is to be interacted with"—undermined?

Position D's Answer: This does not constitute a counterexample. The reasons are as follows:

First, in quantum entanglement, the measurement on A does affect B, but this influence **cannot** be used to transmit information. To confirm the state-change of B, an observer must obtain the measurement result of A via a classical channel (interaction limited by the speed of light) and then compare it with the measurement result of B. The acquisition of information still depends upon interaction.

Second, what quantum entanglement reveals is not "no interaction," but **the non-locality of interaction**. The two particles in an entangled state constitute an indivisible holistic system. The measurement on A is essentially an interaction with this holistic system. The interaction occurs at end A, and its effect manifests simultaneously at end B. This is not "no interaction," but the "holistic character" of interaction. Third, Position D does not stipulate that the type of interaction must be local or classical. It stipulates only that existence must present itself in interaction with other systems. Quantum entanglement reveals a more profound form of interaction, but it does not overturn the principle that "to exist is to be interacted with."

Verification: If quantum entanglement could be used to transmit information faster than light, that would constitute a genuine challenge to Position D. But the fundamental conclusion of quantum mechanics is that it cannot. Therefore, quantum entanglement is not a counterexample, but an extension and confirmation of Position D at the microscopic scale.

Test Conclusion: D passes. It can accommodate the discoveries of quantum mechanics without revising its core claim.

4.4.4 Test Four: Explaining the Origin of the Subject-Object Illusion

Test Design: If Position D is correct—that cognition and existence are primordially co-constituted in interaction—then why do humans so universally and stubbornly fall into the illusion of the "subject-object bifurcation"? A coherent theory must be able to explain how its own "not being discovered" is possible.

Position D's Answer: The illusion of the subject-object bifurcation has two roots, one in the structure of interaction itself, the other in the character of symbolic encapsulation.

Root One: The Asymmetry of Action-Intention. In direct interaction (as when a person picks up a stone), the human, as an intentional agent, experiences "I initiated this action." This unidirectionality of intention constructs, at the level of perception, the asymmetrical experience of "I (subject) act upon the stone (object)." Although physically the interaction between person and stone is reciprocal (Newton's Third Law), the human perceptual system preferentially captures the issuance of intention and sensory feedback, not reciprocity itself.

Root Two: The Subject-Predicate Structure of Symbolic Encapsulation. When humans describe interaction in language, the subject-predicate structure of language—"I pick up the stone"—further

solidifies this asymmetry. "I" becomes the grammatical subject of all actions; "stone" becomes the grammatical object. This grammatical structure is so universal that it in turn shapes the way we understand the world. We come to believe that the world is composed from the start of "subjects" (initiators of actions) and "objects" (recipients of actions). The transparency of symbolic encapsulation—once encapsulation is complete, we operate directly upon the symbols and forget the interaction behind them—allows this illusion to be transmitted from generation to generation.

Verification: Position D not only does not evade this question, but takes it as an important component of its explanatory power. It explains why a correct theory is difficult to discover spontaneously: because certain structural features of interaction (the asymmetry of intention) and certain universal features of symbolic systems (the subject-predicate structure) jointly produce the cognitive inertia of the subject-object bifurcation. A and C cannot explain the origin of this illusion (if the world were already divided into subject and object, it would not be an illusion); if B admits the illusion, it cannot explain how humans can "discover" it to be an illusion (for discovery means stepping outside the symbolic system, and B's framework provides no possibility of stepping outside the symbolic system).

Test Conclusion: D passes. It explains its own "not being discovered," which is itself a manifestation of its theoretical power.

4.5 Formalized Statement: The Birth of Meta-Theorem II

Position D has passed all pressure tests. We now formalize it as a meta-theorem.

Meta-Theorem II: The Interactional Co-constitution Theorem of Cognition and Existence

Dependencies:

- [Meta-Axiom Zero: The Principle of Self-Organizing Tendency] (Existence itself harbors a self-organizing tendency that drives inter-systemic interaction)
- [Meta-Theorem I: The Eternal Dynamical Cycle of Existence] (Existence is the eternal cycle of potential field → system emergence → system evolution → system collapse → return to potential field)
- [Axiom Three: The Principle of Existence and Interaction of Systems] (To exist is to be interacted with)
- [Theorem Three: The Frame-Relativity Theorem of the Observer] (Observation is the joint product of the frameworks of the observing system and the observed system)

Statement:

Existence and cognition are not independent entities, but two inseparable facets of inter-systemic interaction. What we call the "attributes" of existence are patterns stably presented in interaction; what we call "cognition" is the state-change induced within a particular system by interaction. Any dualistic description that separates existence and cognition into "object" and "subject" is a cognitive illusion produced by the operation of symbolic encapsulation, not the constitutive mode of existence itself.

Explication:

This theorem brings to an end the millennia-old debate in the history of philosophy concerning "How does the subject know the object?" Both parties to the debate—realism and constructivism—have taken for granted the presupposition of the "subject-object bifurcation." This theorem shows that this very presupposition is a product of the operation of symbolic encapsulation.

I. The Dissolution of the "Subject-Object Bifurcation"

In direct interaction with the world, humans experience "I act upon the world" because of the

unidirectionality of action-intention; the subject-predicate structure of symbolic systems ("I pick up the stone") further solidifies this experience. When the operation of encapsulation is forgotten, we come to believe that the world is composed from the start of two kinds of entity, "subject" and "object." In truth, subject and object are not primordial forms of existence, but cognitive byproducts of the retrospective encapsulation of the process of interaction.

II. The Reinterpretation of "Existence"

A system "exists" not because it has a "self" stripped of all interactions, but because it can stably present a series of patterns in interaction with other systems. The "hardness" of a stone is the pattern of resistance stably presented in interaction with fingers, hammers, water currents, and other systems. The collection of these patterns is what we call "the stone." There is no "stone itself" stripped of all these patterns.

III. The Reinterpretation of "Cognition"

Cognition is not the "reflection" or "construction" of an object by a subject, but the state-change induced within a particular system by interaction. When a human interacts with a stone, a series of changes occur within the human system—neural activation, muscle memory, emotional association, symbolic marking. The collection of these changes is what we call "cognition of the stone." Cognition is neither prior to interaction (not a priori categories) nor posterior to interaction (not passive reflection). Cognition is the internal facet of interaction.

IV. The Repositioning of the Function of Symbolic Systems

Symbolic systems are neither the mirror of the world (impossible) nor the carver of the world (that is an illusion). Symbolic systems are tools for the encapsulation and transmission of interactional patterns. They compress stable patterns of interaction between humans and the world into a symbol ("hard"), allowing other individuals to share this pattern without repeating all the interactions. The arbitrariness of the symbol (the invented facet) and the stability of the pattern to which the symbol refers (the discovered facet) are here unified.

V. Explaining "Why Did All Predecessors Make This Mistake Collectively?"

Predecessors fell into the subject-object bifurcation because symbolic encapsulation possesses transparency—once encapsulation is complete, we operate directly upon the symbols and forget the interaction behind them. We mistake the logical relations among symbols for the structure of existence itself. This is not a moral failing, but an inevitable cognitive byproduct when symbolic systems become complex enough to self-refer.

Cross-Disciplinary Anchorings:

- **Physics:** The quantum measurement problem—the interaction between the observing system and the observed system jointly defines the presentation of "attributes." There is no "position of the electron" independent of the measuring apparatus.
- **Cognitive Science:** Embodied cognition theory—cognition depends upon sensorimotor interaction between body and environment. Piaget's genetic epistemology shows that the subject-object differentiation is a product, not a premise, of cognitive development.
- **Linguistics:** Saussurean arbitrariness—the relation between signifier and signified is arbitrary, but the interactional pattern to which the signified refers is stable.

- **Philosophy:** Resolves the millennia-old realism-constructivism debate. Each side saw one facet of the duality.

Philosophical Dialogue—With Kant:

Kant saw correctly that cognition requires categories, but he wrongly inferred that "the thing-in-itself is unknowable." In the perspective of this theorem, "the thing-in-itself" is a pseudo-concept—it is the empty space left when one strips the interactional pattern away from the process of interaction. Existence is not something hiding behind phenomena; existence is the stable presentation of interactional patterns themselves. The metaphor of the glass (categories as glass through which we see the world) is itself an illusion produced by symbolic encapsulation.

Chapter 5: Discussion—Dialogue with the Ancestors

The value of a meta-theorem lies not only in its internal logical coherence, but also in its capacity to enter into dialogue with the deepest insights in the history of thought. This chapter situates Meta-Theorem II within the coordinate system of the history of philosophy, engaging in deep dialogue with three ancestors: Kant, Saussure, and Wittgenstein. The aim of this dialogue is not to "defeat the predecessors," but to locate with precision: What truths did they see? Where did they stop? What does Meta-Theorem II inherit, what does it revise, and where does it transcend?

5.1 Dialogue with Kant: The Dissolution of the Thing-in-Itself

Kant is among those who came closest to Meta-Theorem II in the history of philosophy. His *Critique of Pure Reason* effected a revolutionary reversal: it is not cognition that conforms to objects, but objects that must conform to our cognitive conditions. This reversal is customarily called the "Copernican Revolution."

The Truth Kant Saw

Kant saw correctly that human cognition of the world must proceed through certain fundamental categories—time, space, causality, and so on. These categories are neither derived from experience (contra empiricism) nor arbitrary conventions (contra skepticism), but are the a priori structures of the human cognitive subject. Without these categories, experience itself would be impossible. In this sense, cognition is not passive reflection, but active construction. Kant provided a foundation for the universal necessity of knowledge that does not depend upon "existence itself"—it is grounded in the a priori structures of the subject shared by all humans.

Where Kant Stopped

But Kant stopped at a crucial step. He insisted that the a priori categories are only structures of the human subject, unrelated to the "thing-in-itself." The thing-in-itself exists, but it is unknowable. We can only know phenomena—the empirical world organized by a priori categories—and can never reach the world itself as it is prior to this organization.

This "thing-in-itself" became the most fragile link in Kant's system. The subsequent German Idealists (Fichte, Schelling, Hegel) saw acutely: if the thing-in-itself is unknowable, how do we even know it exists? If it exists and exerts some effect upon cognition, why can it not be cognized? Kant himself could not consistently maintain his position—on the one hand, he says the thing-in-itself stimulates the senses

to produce the material of experience; on the other hand, he says the thing-in-itself is unknowable. If it can "stimulate," then it has already interacted with our senses, and interaction necessarily presents some pattern. Why is this pattern not cognition?

Meta-Theorem II's Response

In the perspective of Meta-Theorem II, the "thing-in-itself" is a pseudo-concept.

Kant presupposed a dichotomy between "phenomena" and the "thing-in-itself." This very dichotomy is a refined version of the illusion of the subject-object bifurcation produced by the operation of symbolic encapsulation. Kant assigned "cognitive conditions" (a priori categories) to the subject, assigned "what is cognized" to phenomena, and then posited a "thing-in-itself" stripped of all cognitive conditions as the source of phenomena. But he never asked: is this very positing of a "source" itself a cognitive operation? If it is, then the "thing-in-itself" is already within cognition; if it is not, then on what grounds can we posit it?

Meta-Theorem II shows: there is no thing-in-itself "behind" phenomena. Existence is precisely the stable presentation of patterns in inter-systemic interaction. The hardness of a stone is neither an attribute of the "stone itself" nor a label imposed by humans upon the stone, but a pattern stably presented when the stone interacts with human fingers. This pattern belongs neither to the stone alone nor to the human alone; it is the product of interaction. There is no stone "prior to interaction," nor is there human cognition "prior to interaction." Both co-present within a single current of interaction.

Kant asked: "How can we know the thing-in-itself?" Meta-Theorem II answers: "The question is wrongly posed. The question is not 'How do we know the thing-in-itself?' but 'How do interactional patterns present themselves?' The 'thing-in-itself' is the empty space left when you strip interaction away from cognition. That empty space is not existence itself, but a trace left by your operation of abstraction."

Inheritance and Transcendence

Meta-Theorem II inherits Kant's "Copernican Revolution"—cognition is not passive reflection. But it transcends Kant's negative conclusion that "the thing-in-itself is unknowable." It shows that cognition can guide action not because it has a mysterious correspondence with the thing-in-itself, but because cognition is itself the product of interaction. Airplanes fly not because we have discovered the thing-in-itself of "flight," but because we have stably presented the pattern of "flight" in countless interactions with air, materials, and energy. This pattern is real—not because it reflects the thing-in-itself, but because it is stably efficacious in interaction.

5.2 Dialogue with Saussure: The Two Facets of Arbitrariness

Saussure's structural linguistics is one of the most important sources of the constructivist current. His core insight is that the meaning of a sign does not come from correspondence to external things, but from relations of difference within the sign system. The sign "cat" has meaning not because it refers to that furry animal, but because it is not "dog," not "mouse," not "tiger." Meaning is the product of internal differentiation within the system.

The Truth Saussure Saw

Saussure saw correctly that the relation between signifier (sound-image) and signified (concept) is arbitrary. The same animal is called *māo* in Chinese, *cat* in English, *chat* in French. There is no intrinsic reason why it must be called one sound rather than another. The value of a sign is determined by its

position within the entire sign system, not by its correspondence to an external thing. This insight profoundly unsettled the naive realism that takes language to be a picture of the world.

Where Saussure Stopped

But Saussure (and the constructivist tradition influenced by him) ran into difficulty on one issue: if the meaning of a sign is entirely determined by internal differences within the system, why is translation between different sign systems possible? If the meaning of "cat" is merely "not dog, not mouse...", then why do Chinese *māo* and English *cat* roughly correspond? Saussure's answer was that different linguistic systems carve up the world in different ways. But he could not explain why the differences in carving are not unlimited—why do all languages distinguish "animal" from "stone," "alive" from "dead," and all possess expressions resembling "causality"?

Saussure's principle of arbitrariness saw only the arbitrariness of the sign's form (the relation between signifier and signified), but did not ask: is the signified itself arbitrary? Is the signified "cat"—that concept of a furry four-legged animal—an arbitrary human construction, or a pattern stably presented in the long-term interaction between humans and a certain kind of animal?

Meta-Theorem II's Response

Meta-Theorem II provides a more complete framework for Saussure's principle of arbitrariness: symbolic systems possess a duality—the discovered facet and the invented facet.

The invented facet corresponds to the arbitrariness Saussure saw. The relation between the sign's form (signifier) and the signified concept is arbitrary and conventional. Different languages can use entirely different sounds to refer to the same animal. Logical symbol systems can evolve from Aristotelian syllogistic to modern predicate logic; the forms of symbols can change in countless ways.

The discovered facet corresponds to the side Saussure missed. The interactional pattern to which the signified refers is not arbitrary. The signified of the symbol "cat" is the collection of patterns stably presented in long-term interaction between humans and that furry, meowing, mouse-catching animal. The stability of this collection of patterns does not depend upon what symbol we use to mark it. Primitive humans, without the word "cat," could already recognize the pattern of this animal in interaction (it moves, it catches things, it must be avoided or domesticated).

Saussure's error was not that he saw arbitrariness, but that he took arbitrariness to be the whole of the symbolic system. Meta-Theorem II shows: the invented facet of the symbol is arbitrary, but the discovered facet—the interactional pattern it encapsulates—is stable and cross-cultural. This explains why translation across different languages is roughly possible: not because they share the same symbolic system, but because their symbolic systems encapsulate roughly the same interactional patterns. Human interaction with cats presents similar patterns in all cultures, so the signified of "cat" roughly coincides across cultures.

Inheritance and Transcendence

Meta-Theorem II inherits Saussure's insight of arbitrariness (the invented facet), but supplements the missing half (the discovered facet). It explains why language possesses both the freedom of human convention (the invented facet) and cross-cultural constraint (the discovered facet). Language is not the carver of the world—carving is an illusion, as if the world were originally continuous and language sliced it into discrete chunks. Language is the encapsulator of interactional patterns—the patterns are already stably presented in interaction; language merely attaches a label to them for transmission within the group.

5.3 Dialogue with Wittgenstein: What Cannot Be Said, Can Be Said

Wittgenstein is the most resolute "delimiter" in the history of philosophy. He declared in the *Tractatus Logico-Philosophicus*: *Whereof one cannot speak, thereof one must be silent*. What sorts of things cannot be said? Ethics, aesthetics, the meaning of life, metaphysics, and "the feeling of the world as a limited whole." The task of philosophy is not to say these things, but to clarify language so that these things may "show" themselves, after which philosophy is to be thrown away like a ladder.

The Truth Wittgenstein Saw

Wittgenstein saw correctly that language has a boundary. In his *Tractatus* period, he held that language can only describe "facts"—possible states of affairs in logical space. Any attempt to describe something "beyond facts" results in nonsense. In his later *Philosophical Investigations*, he turned to the view that the meaning of language lies in its use, and that philosophical illness arises from language going on holiday—words are lifted out of their everyday language-games and spin idly in abstraction. In both periods, he insisted that the task of philosophy is not to construct theories, but to provide therapy—to bring words back from their metaphysical flight to the everyday ground.

The profundity of Wittgenstein lies in his acute identification that when language attempts to speak of itself, of the whole, of foundations, it falls into paradox and idle spinning. This "self-reference" is the forbidden zone of language. His critique of "private language" and his exposure of the "rule-following" paradox both point to the same core: the meaning of language cannot be guaranteed by private inner ostensive definition, but must rest upon public, verifiable language-games.

Where Wittgenstein Stopped

But Wittgenstein wrongly inferred "existence has a boundary" from "language has a boundary." He took the "limit of linguistic expression" he discovered to be the "ineffability of existence." The things he forbade philosophy to talk about—ethics, meaning, existence itself—are not, in his view, non-existent, but "unsayable." Yet "unsayable" is itself already a saying. He said "Whereof one cannot speak, thereof one must be silent," but this very sentence is a saying of that whereof one cannot speak. Wittgenstein himself could not obey his own prohibition.

A more fundamental problem: if language is merely language-games, and language-games are rooted in forms of life, then where do forms of life themselves come from? Why do human forms of life exhibit rough cross-cultural similarity? Why do all cultures have a language-game of "causality"? Wittgenstein refused to answer these questions, because answering them would require stepping beyond language-games to speak of that which makes language-games possible. And this is precisely what his prohibition forbids.

Meta-Theorem II's Response

In the perspective of Meta-Theorem II, Wittgenstein's "unsayable" is not the boundary of existence, but the boundary of the particular linguistic tool he used—referential language.

Wittgenstein's argument implicitly depends upon a presupposition: saying is "describing facts with propositions." When he found that certain things (ethics, meaning, existence as a whole) cannot be described in this linguistic form, he concluded that they are "unsayable." But he did not ask: is the linguistic tool we are using too narrow? If we switch to another linguistic tool—one that describes not static facts, but dynamical processes, interactional patterns, inter-systemic relations—can those

"unsayable" things then be legitimately said?

Meta-Theorem II shows that existence itself is not a collection of "facts," but a dynamical process of inter-systemic interaction. The regularities of this process (self-organizing tendency, systemic constraint) can be described in dynamical language, without appeal to Wittgensteinian "reference." When we speak of "self-organizing tendency," we are not referring to some mysterious entity, but describing an observable, cross-level dynamical pattern—one sees its traces everywhere, from the condensation of nebulae to the evolution of life to the rise of civilizations. This description does not violate Wittgenstein's requirement of "language-games"—it is anchored in publicly verifiable interactional patterns.

Wittgenstein said: "What cannot be said can only be shown." Meta-Theorem II responds: "What you call 'showing' is in fact 'presenting in interaction.' And 'presenting in interaction' is precisely the definition of 'existence' given by Meta-Theorem II. Therefore, showing is the mode of existence, and the mode of existence can be said in dynamical language. Your prohibition stems from allowing only one linguistic tool—referential language—into philosophy. Switch tools, and the prohibition is naturally lifted."

Inheritance and Transcendence

Meta-Theorem II inherits Wittgenstein's sensitivity to the boundaries of language—not every linguistic form is suited to speak of existence. But it transcends Wittgenstein's prohibition—it has invented a linguistic form suited to speaking of existence: the language of system dynamics. This language does not describe "facts," but "the stable presentation of interactional patterns." It does not refer to "mysterious entities," but tracks "cross-level dynamical patterns." What was "unsayable" in Wittgenstein—ethics, meaning, existence—becomes, in this new language, a dynamical description that can be publicly verified. Philosophy turns from "therapy" back to "discovery."

5.4 Chapter Conclusion: The Philosophical-Historical Coordinates of Meta-Theorem II

The dialogue with Kant dissolves the "thing-in-itself"—the empty space left after stripping away interaction.

The dialogue with Saussure distinguishes the invented and discovered facets of the symbol—what is arbitrary is the symbol's form; what is stable is the interactional pattern.

The dialogue with Wittgenstein dismantles the prohibition of the "unsayable"—it is not existence that is unsayable, but referential language that is unsuited to say it. Switch languages, and existence can be legitimately said.

These three dialogues jointly delineate the philosophical-historical coordinates of Meta-Theorem II: it is the naturalization of Kant's transcendental philosophy—taking the "a priori categories" out of the human subject and placing them back into inter-systemic interaction; it is the completion of Saussure's principle of arbitrariness—demarcating its boundary (the invented facet is arbitrary, the discovered facet is stable); it is the lifting of Wittgenstein's prohibition—taking what was "unsayable" and making it legitimately sayable in dynamical language.

Meta-Theorem II does not negate the predecessors. It completes them—taking the truths each saw and placing them within a more complete framework, allowing each to find its proper place.

Now we may proceed to the final conclusion.

Chapter 6: Conclusion—Contributions, Significance, and Openness

The discovery process of a meta-theorem has now been presented in full. What remains is to clarify:

What new thing does this discovery provide? Where does it point? And what are its boundaries?

6.1 The Threefold Contribution of This Paper

The contribution of this paper is not the proposal of a "new view"—the history of philosophy has never lacked new views. Its contribution lies in having derived, by means of a publicly verifiable method, a meta-theorem from boundary crises, and having endowed it with precise formalized statement. Specifically, this paper makes a threefold contribution.

First: A Theorematic Contribution.

This paper has discovered and formalized the "Interactional Co-constitution Theorem of Cognition and Existence" (Meta-Theorem II of System Ontology). This theorem asserts: existence and cognition are not independent entities, but two inseparable facets of inter-systemic interaction. What we call the "attributes" of existence are patterns stably presented in interaction; what we call "cognition" is the state-change induced within a particular system by interaction. The subject-object bifurcation is a cognitive illusion produced by the operation of symbolic encapsulation.

This theorem brings to an end the millennia-old debate in the history of philosophy concerning "How does the subject know the object?" It shows that the parties to the debate—realism, constructivism, transcendental philosophy, and the sprouts of process realism—all shared the unexamined presupposition of the subject-object bifurcation. Meta-Theorem II dissolves this presupposition, thereby dissolving the original question. It is not a "middle way" between realism and constructivism, but a leap out of their shared plane.

Second: A Methodological Contribution.

This paper has presented a complete operational demonstration of Boundary-Driven Abductive Locking. Proceeding from a boundary crisis (the realism-constructivism impasse), executing a logical reversal, delineating a possibility space, establishing criteria and excluding positions one by one, applying pressure tests, and finally formalizing—each step is open to the reader's public scrutiny.

The value of this demonstration is this: it transforms meta-theoretical discovery from the private affair of genius into a publicly operable and verifiable procedure. The reader may disagree with the conclusion of this paper, but if they wish to refute it, they must point to a specific flaw in the operations—an omission in the possibility space? An unfair criterion? A loose exclusion? A failed pressure test? Abductive Locking does not claim to have found absolute truth; it claims only that, given the operations performed, the position that remains is the most coherent, most parsimonious, and most explanatorily powerful answer available under current cognitive conditions.

Third: A Paradigmatic Contribution.

This paper furnishes a new ontological foundation for philosophy, cognitive science, and civilizational diagnostics.

For philosophy, Meta-Theorem II dissolves the millennia-old realism-constructivism debate and provides a new starting point for epistemology and metaphysics that does not depend upon the subject-object bifurcation.

For science, it offers a unified ontological interpretation of the quantum measurement problem (the interaction between observing system and observed system jointly defines the presentation of attributes), embodied cognition theory (cognition depends upon sensorimotor interaction between body and

environment), and linguistic arbitrariness (what is arbitrary is the symbol's form; what is stable is the interactional pattern).

For civilizational diagnostics, it provides the ultimate metaphysical root for understanding a series of cognitive pathologies. Nihilism—mistaking the inability of symbolic systems to self-justify ultimate meaning for the directionlessness of existence itself. Conceptual suspension—the loss of anchorage to interactional patterns in the intergenerational transmission of academic terminology. Academic jargon—the closed game produced when symbolic encapsulation forgets its own instrumentality. These phenomena are all symptoms of the "subject-object illusion" in different domains.

6.2 The Boundaries and Openness of This Paper

An honest theorem must clearly state its boundaries. Meta-Theorem II has three boundaries.

First, it is an ontological theorem, not an empirical-scientific theorem. It does not directly predict any concrete empirical phenomena; it only provides the underlying framework for understanding empirical phenomena. It answers "What is the relation between cognition and existence?" not "How does some concrete cognitive process operate?" The latter is the task of cognitive science and neuroscience. Meta-Theorem II provides an ontological foundation for such empirical research, but does not replace it.

Second, its method of discovery—Boundary-Driven Abductive Locking—depends upon the exhaustion of the possibility space and the establishment of criteria. If future thinkers discover an omission in this paper's possibility space (a fifth logically possible position exists), or find that the criteria require adjustment (a more fundamental criterion was overlooked), then the locking can be reopened and the conclusion revised. The spirit of Abductive Locking is not to lock down truth once and for all, but to approach, continuously and rigorously, the answer that is most coherent under current cognitive conditions.

Third, Meta-Theorem II itself obeys the law it reveals. It is the product of the interaction between the author (a cognitive system) and the problem of "the relation between cognition and existence" (an existence system). Its discovered facet (the regularity stated by Meta-Theorem II) and its invented facet (this paper's symbolic organization, case selection, argumentative structure) constitute a duality. This means that Meta-Theorem II is not "the final truth," but "the most coherent answer under current cognitive conditions." If a better answer emerges in the future—whether due to new facts, new tools, or a new logical reversal—Meta-Theorem II should be revised or replaced. It itself reserves the possibility of being transcended.

6.3 Pointing Toward the Future

The discovery of Meta-Theorem II is not an endpoint, but a beginning. It opens a series of further inquiries.

First, how is Meta-Theorem II to be applied in concrete domains? This paper has only completed the formalization of the meta-theorem itself. Its concrete unfolding in cognitive science, linguistics, physics, and sociology awaits subsequent research. For example: Can Piaget's genetic epistemology be reformulated in terms of Meta-Theorem II? What new aspect does Saussurean semiology present within the framework of duality? Can the quantum measurement problem be redescribed in the language of interactional co-constitution?

Second, can Boundary-Driven Abductive Locking be used to discover other meta-theorems? This paper has only demonstrated the application of Abductive Locking to a single case (the relation between cognition and existence). Can this method be used to discover meta-theorems concerning "value,"

"consciousness," "freedom," and other problems? If so, what adjustments must be made to its operational steps? Can Abductive Locking itself be formalized as a more general algorithm for meta-theoretical discovery?

Third, what does Meta-Theorem II imply for civilizational diagnostics? If the subject-object bifurcation is an illusion of symbolic encapsulation, can the evolution of civilization be understood as a cognitive phase transition from "immersion in illusion" through "recognition of illusion" to "transcendence of illusion"? What are the conditions for this phase transition? What is its relation to the "meta-cognitive collective intelligence" proposed in *Civilization Dynamics*? These questions cannot be unfolded within this paper, but they are precisely the future toward which Meta-Theorem II points.

6.4 A Final Resonance

Stone is hard. This is true in every known human culture.

But we can finally say clearly what this "truth" actually means.

It does not mean that the stone "itself" possesses an attribute called "hardness," which humans then discovered. It means that humans, in interaction with stone, stably experience a pattern of resistance, and then use the symbol "hard" to encapsulate this pattern for transmission and coordination within the group. The stone is real. The interaction is real. The pattern is real. The symbol is a tool. Mistaking the symbol for existence itself is an illusion.

The realists saw the stability of the pattern, and they were right. The constructivists saw the mediating role of the symbol, and they were right. Their error was not in seeing wrongly, but in taking the part for the whole, in taking one facet of the duality for the entirety of the duality.

Meta-Theorem II does not aim to defeat them. It aims to complete them—to take the truths each saw and place them within a more complete framework, allowing each to find its proper place.

Perhaps this is what philosophy can ultimately offer: not a statue of truth to be worshipped, but a cognitive scalpel that can be continuously sharpened.

And this scalpel is now placed in your hands.

Chapter 7: Reflexive Verification—How This Paper Obeys Its Own Theorem

A meta-theorem that claims to reveal a universal relation between cognition and existence must be able to explain how it itself was written. If the writing process of this paper violates Meta-Theorem II, then this paper is self-refuting; if the writing process of this paper is itself an instance of Meta-Theorem II, then the form and content of this paper constitute a unity.

The task of this chapter is to complete this reflexive verification.

7.1 The Writing of This Paper Is an Interaction

This paper did not emerge ex nihilo from the "pure speculation" of the author. It is the product of the long-term interaction between the author (a cognitive system) and the problem of "the relation between cognition and existence" (an existence system).

This interaction can be traced back quite far. The author's experiences on construction sites, conversations with colleagues about "the rules," the felt sense of the alienation of academic institutions, the intellectual struggle with nihilism—these are not "pure thinking," but the author, as a human being, interacting with concrete social systems, symbolic systems, and material systems. In these interactions, certain patterns repeatedly presented themselves: How are rules tacitly obeyed? How does authority define "right and

wrong"? How is individual cognition shaped by systems? These patterns constitute the raw material of Meta-Theorem II.

The writing of this paper is the systematization and formalization, through Boundary-Driven Abductive Locking, of the patterns stably presented in these long-term interactions, ultimately encapsulated in the symbolic form of Meta-Theorem II. In this process, the author is not a "pure subject," nor is this paper a "pure object." Author and paper co-present within a single current of writerly interaction. The author writes, but what is written in turn shapes the author—each round of revision is a renewed interaction between the author and the already-written text, from which more precise expression emerges.

7.2 The Duality of This Paper: The Discovered and the Invented

Meta-Theorem II shows that any cognitive product possesses a duality—the discovered facet (the regularity itself) and the invented facet (the symbolic representation). This paper is no exception.

The discovered facet of this paper is the regularity revealed by Meta-Theorem II—that cognition and existence co-present in interaction. This regularity was not "invented" by the author. It is the most coherent position remaining after the other logical possibilities were excluded step by step, through Boundary-Driven Abductive Locking, from cross-disciplinary boundary crises. The author did not "create" this regularity; the author merely extracted it from phenomena and endowed it with clear form. In this sense, it was discovered.

The invented facet of this paper is the symbolic organization invented by the author to convey this regularity. This includes: beginning from the everyday puzzlement of the stone's hardness (rather than from the literature); adopting the five-step structure of Abductive Locking; marking the possibility space with "Position A, B, C, D"; designing the stone thought experiment, the interaction-type analysis, the quantum entanglement test, and the other pressure tests; using the term "interactional co-constitution"; and the hybrid classical-modern genre of this paper itself. These modes of symbolic organization are not logically necessary. The author could have chosen a different entry point, different cases, different terms, a different structure. The author chose the present form because it was judged, in the anticipated interaction with the reader, to be the most likely to effectively convey the discovered facet. In this sense, it was invented.

The discovered and the invented are inseparable. Without the discovered facet, this paper would be mere rhetoric; without the invented facet, the discovered could not enter public cognition. This paper can exist as a paper precisely because it possesses both a discovered facet (Meta-Theorem II) and an invented facet (the symbolic organization of this paper), and because the two facets are mutually dependent.

7.3 The Source of This Paper's Legitimacy: External Judgment

A meta-theoretical paper faces a classic dilemma: if it claims to reveal the law of "how cognition is possible," whence comes the cognitive legitimacy of the paper itself? It cannot justify itself by itself—that would be circular. It cannot appeal to a higher authority—that would lead to infinite regress.

The solution of this paper, already foreshadowed in the introduction and implicit in the formalized statement of Chapter 4, can now be stated explicitly: the legitimacy of this paper is not established by circular self-justification within the system, but is adjudicated by the reader as an external observer.

This is itself an application of Meta-Theorem II. Meta-Theorem II shows that cognition is the product of inter-systemic interaction, and that the legitimacy of cognition comes not from an a priori guarantee internal to the "subject," but from whether the pattern stably presented in interaction can pass public scrutiny.

The reader is another cognitive system. The reader interacts with this paper—reading, understanding, questioning, verifying. In this interaction, whether the discovered facet of this paper (the regularity revealed by Meta-Theorem II) can be stably presented within the reader's cognitive system depends upon whether the invented facet (the symbolic organization of this paper) is sufficiently clear, coherent, and explanatorily powerful. If, after reading, the reader can independently reconstruct the chain of reasoning of this paper and test it against their own experience, then this paper has accomplished its cognitive transmission. If, in reading, the reader discovers a flaw—an omission in the possibility space, an unfair criterion, a loose exclusion, a failed pressure test—then the legitimacy of this paper is weakened or negated.

This power of judgment lies not in the hands of the author, but in the hands of the reader. What the author can do is only to present the reasoning process as clearly and honestly as possible, laying every step open to public scrutiny. This is precisely the spirit of Boundary-Driven Abductive Locking, and the praxis of Meta-Theorem II in writing.

7.4 The Closure Is Complete

This paper has now completed its self-referential closure.

This paper discovered Meta-Theorem II—that cognition and existence co-present in interaction—through Boundary-Driven Abductive Locking. The writing process of this paper is itself the product of the interaction between the author (a cognitive system) and the problem (an existence system). The discovered facet (Meta-Theorem II) and the invented facet (the symbolic organization of this paper) constitute a duality. The legitimacy of this paper is not self-declared by the author, but adjudicated by the reader in the interaction of reading.

A meta-theory that claims to reveal the essence of cognition can explain how it itself was written. This is not only the coherence of the theory, but the honesty of the theory.

The completion of this closure is not the endpoint of this paper, but the starting point of the reader's verification.

Now, turn back to the front matter and re-examine the reading guide. What this paper invites you to do has been the same all along: serve as an independent rational arbiter, follow each step of the reasoning, and scrutinize the soundness of each step. If you discover a flaw, point it out. If you find a better answer, speak it.

This cognitive scalpel is now placed in your hands.

Further Reading and Academic Background Notes

I. The Theoretical Foundation of This Paper

This paper is a meta-theoretical work within the author's "Noetic Trilogy." It assumes some familiarity on the reader's part with the basic framework of System Ontology, though this is not strictly necessary—the core argument of this paper (Chapter 4) is as self-contained as possible. To facilitate deeper understanding of the theoretical foundations of this paper and to explore the further questions it opens, these notes on further reading and background are provided.

1. Core Axioms of System Ontology

The axioms of System Ontology upon which this paper depends have been noted throughout the main text. They are collected here for the reader's convenience.

Meta-Axiom Zero: The Principle of Self-Organizing Tendency—Existence itself harbors an eternal, non-teleological, far-from-thermodynamic-equilibrium tendency toward self-organization. This tendency is the ultimate driving force by which systems emerge from the potential field and maintain their ordered structures.

Meta-Axiom One: The Principle of Systemic Constraint—Any system that can be stably articulated and reasoned about must contain within itself a set of self-constraining meta-rules that dissolve the legitimacy of "vicious self-reference" that would otherwise lead to logical collapse, thereby guaranteeing the system's stability and intelligibility.

Meta-Theorem I: The Eternal Dynamical Cycle of Existence—Potential Field → System Emergence → System Evolution → System Collapse → Return to Potential Field. This cycle proceeds in nested fashion across countless levels, without beginning, end, or external purpose.

Axiom Three: The Principle of Existence and Interaction of Systems—All that can be articulated, reasoned about, and interacted with is a "system." The sole proof and sole mode of a system's existence is its interaction with other systems. A system that interacts with nothing is equivalent to non-existence.

Theorem Three: The Frame-Relativity Theorem of the Observer—Any observation of a system is essentially an interaction between the frameworks of the observing system and the observed system. Therefore, all observational results are the joint product of two (or more) systems. There is no God's-eye "absolute objectivity."

For the complete deduction, cross-disciplinary anchorings, and philosophical dialogues of the above axioms and theorems, see the author's monograph *System Ontology: The Ultimate Jurisprudence of Existence*.

2. The Complete Statement of Boundary-Driven Abductive Locking

Chapter 3 of this paper outlined the five steps of Abductive Locking; Chapter 4 demonstrated its operation. For the complete formalized statement of this method—including its epistemological status, its dialogue with traditional methods of discovery, and case studies of its application in different domains—see the Introduction and Appendix of *System Ontology* ("Methodological Supplement: Boundary-Driven Abductive Locking").

3. Further Unfolding of Meta-Theorem II

Meta-Theorem II (the Interactional Co-constitution Theorem of Cognition and Existence), discovered in this paper, is the pivot connecting System Ontology (the meta-theoretical layer) with Noetic Ecology (the

applied layer). It unfolds in the following directions.

Cognitive Science Direction: Meta-Theorem II provides an ontological foundation for embodied cognition and genetic epistemology. Piaget's concept of "sensorimotor schemata" and Merleau-Ponty's concept of the "body-subject" can be reinterpreted within this framework as "the internalization of interactional patterns." For related discussion, see Theorem E25 (The Synergistic Resonance and Stratified Model of Conscious Phenomena) and Theorem D23 (The Cognitive Escape Theorem) in *Noetic Ecology*.

Linguistics Direction: The dual-aspect reinterpretation of Saussure's principle of arbitrariness (the invented facet is arbitrary; the discovered facet is stable) is fully unfolded in Theorem C15 (The Duality Theorem of Logic) of *Noetic Ecology* and in the paper *The Duality of Logic: The Unity of Discovery and Invention*.

Civilizational Diagnostics Direction: Once the subject-object illusion is revealed as a cognitive byproduct of symbolic encapsulation, a series of cognitive pathologies at the civilizational scale—nihilism, conceptual suspension, academic jargon, the cognitive rupture between experts and the public—acquire a unified metaphysical root. For related discussion, see *A Critique of Academic Jargon: The Alienation and Antagonism of Modern Knowledge Production* and *The Absence of Antagonism: System Dynamics and the Cognitive Roots of Civilizational Imbalance*.

II. Guidance for Deep Reading of the Dialogue with the Ancestors

Chapter 5 of this paper engaged in deep dialogue with Kant, Saussure, and Wittgenstein. Suggested reading pathways are provided here for readers interested in pursuing the philosophical-historical background.

1. Concerning Kant

This paper's dialogue with Kant focuses on the "Transcendental Deduction" and the concept of the "thing-in-itself" in the *Critique of Pure Reason*. Suggested reading:

Kant, I. (1998). *Critique of Pure Reason* (P. Guyer & A. Wood, Trans.). Cambridge University Press. Focus: Preface to the Second Edition (the declaration of the "Copernican Revolution"), the Transcendental Aesthetic (space and time as a priori forms of intuition), the Transcendental Analytic (the Transcendental Deduction of the Categories), and the distinction between Phenomena and Noumena.

For the critical history of the "thing-in-itself" problem, see further: Fichte's *Foundations of the Science of Knowledge* (the first systematic dissolution of the thing-in-itself), Hegel's Introduction to the *Phenomenology of Spirit* (a critique of Kant's thing-in-itself), and P. F. Strawson's *The Bounds of Sense* (a modern analytic reading of Kant).

The gist of this paper's response to Kant: The thing-in-itself is not existence itself, but the empty space left after stripping the interactional pattern away from the process of interaction.

2. Concerning Saussure

This paper's dialogue with Saussure focuses on the "principle of arbitrariness" and the concept of "linguistic value" in the *Course in General Linguistics*. Suggested reading:

Saussure, F. de. (2011). *Course in General Linguistics* (W. Baskin, Trans.). Columbia University Press. Focus: Part One, Chapter I (the arbitrariness of the sign), Part Two, Chapter IV (linguistic value), and the distinction between the synchronic and the diachronic.

For subsequent debates on Saussure's principle of arbitrariness, see further: Benveniste's *Problems in*

General Linguistics (a revision of arbitrariness), Jakobson's *Fundamentals of Language* (the development of structuralist linguistics), and Derrida's *Of Grammatology* (a deconstruction of Saussure's "phonocentrism").

The gist of this paper's response to Saussure: What is arbitrary is the form of the symbol (the relation between signifier and signified); what is stable is the interactional pattern to which the symbol refers. Saussure saw the invented facet and overlooked the discovered facet.

3. Concerning Wittgenstein

This paper's dialogue with Wittgenstein engages both the early *Tractatus Logico-Philosophicus* and the later *Philosophical Investigations*. Suggested reading:

Wittgenstein, L. (1922). *Tractatus Logico-Philosophicus* (C. K. Ogden, Trans.). Routledge & Kegan Paul. Focus: The delimitation of the sayable and the unsayable (from 6.4 onward), and the metaphor of philosophy as a ladder (6.54).

Wittgenstein, L. (1953). *Philosophical Investigations* (G. E. M. Anscombe, Trans.). Blackwell. Focus: Language-games and forms of life (§§1–38), the rule-following paradox (§§185–242), and philosophy as therapy (§§109–133).

On the continuity and discontinuity between the early and later Wittgenstein, see further: Anscombe's *An Introduction to Wittgenstein's Tractatus*, Kripke's *Wittgenstein on Rules and Private Language* (an influential reading of the rule-following paradox), and Baker and Hacker's *Wittgenstein: Understanding and Meaning*.

The gist of this paper's response to Wittgenstein: The "boundary of language" Wittgenstein discovered is the boundary of the referential language he employed, not the boundary of existence. Switch to a different linguistic tool—the language of system dynamics—and what was "unsayable" (ethics, meaning, existence itself) can be legitimately said.

4. Concerning Piaget and Merleau-Ponty

Chapter 2 of this paper positioned Piaget and Merleau-Ponty as "sprouts of process realism." Suggested reading:

Piaget, J. (1970). *The Principles of Genetic Epistemology* (W. Mays, Trans.). Basic Books. Focus: The differentiation of subject and object in the sensorimotor stage, the origins of operativity.

Merleau-Ponty, M. (1962). *Phenomenology of Perception* (C. Smith, Trans.). Routledge. Focus: Introduction (the double critique of empiricism and intellectualism), Part One: "The Body" (the body as the pivot of "being-in-the-world").

For a deep dialogue between these thinkers and System Ontology, see the explication of Theorem E25 in *Noetic Ecology* and its Appendix "Dialogue with Genetic Epistemology."

III. Terminology: Origins and Comparisons

Several core terms used in this paper have origins in the history of philosophy, but this paper endows them with specific technical meanings. Brief origins and comparisons are provided here to help the reader distinguish everyday usage, historical philosophical usage, and the usage specific to this paper.

1. Abduction

Peirce introduced abduction in the 19th century as a third mode of reasoning alongside deduction and induction. Peirce's classic formulation: "The surprising fact, C, is observed; But if A were true, C would

be a matter of course; Hence, there is reason to suspect that A is true."

The Abductive Locking of this paper inherits the basic structure of Peircean abduction (inferring the optimal explanation backward from phenomena), but upgrades it into a rigorous, stepwise, publicly verifiable locking procedure. Peircean abduction remains at the level of "insight"; this paper's Abductive Locking operationalizes insight.

2. Interaction

"Interaction" is a common term both in everyday language and in the history of philosophy. In Kant, interaction is one of the categories of relation; in physics, interaction refers to the four fundamental forces. The "interaction" of this paper is closer to the usage in systems theory, but anchored in Axiom Three of System Ontology: the sole proof of a system's existence is its interaction with other systems. It does not presuppose the type of interaction (which may be physical, informational, or symbolic); it stipulates only that existence must present itself in interaction. Meta-Theorem II redefines "cognition" and "existence" as two facets of interaction.

3. Co-constitution

"Co-constitution" is not a standard philosophical term. This paper uses it to translate *co-constitution*—that cognition and existence are jointly constituted and jointly presented in interaction. Its difference from "construction" lies in this: construction suggests one side active and the other passive (the subject constructs the object); co-constitution emphasizes bidirectionality—cognition and existence have no priority, only mutual dependence. It has an affinity with "emergence," but co-constitution emphasizes simultaneous bidirectionality, whereas emergence emphasizes the irreducibility of the higher level to the lower.

4. Symbolic Encapsulation

"Encapsulation" is borrowed from computer science. In object-oriented programming, encapsulation refers to bundling data with the methods that operate on that data, hiding internal details and exposing only the necessary interface.

This paper uses "symbolic encapsulation" to describe the operation by which humans compress, name, and transmit stable interactional patterns using symbols. The benefit of encapsulation is efficiency—patterns can be shared without repeating all the interactions. The cost of encapsulation is the illusion of transparency—once encapsulation is complete, people tend to operate directly on the symbols, forgetting the interactions behind them. This concept is key to understanding the "subject-object illusion."

5. The Discovered and the Invented

This pair of terms is the core innovation of this paper. They correspond to the application of Axiom Two of System Ontology (Matter/Noetic Duality) to the domain of cognition. The discovered facet refers to that aspect of cognition anchored in interactional patterns—it is stable, cross-cultural, and independent of symbolic form. The invented facet refers to that aspect of cognition shaped by symbolic systems, cognitive frameworks, and historical paths—it is variable, optimizable, and evolvable. This pair of terms resolves the millennia-old realism-constructivism debate: realism saw the discovered facet, constructivism saw the invented facet; each saw one facet of the duality.

IV. Suggested Reading Pathways

Three reading pathways are offered according to the reader's academic background and interests.

Pathway One: Philosophy Readers

For readers interested in the history of philosophy and metaphysics, begin with the philosophical dialogue of Meta-Theorem II (Chapter 5), then trace back to the original works of Kant, Saussure, and Wittgenstein, then enter the meta-axioms and system axioms of *System Ontology*, and finally take *The Duality of Logic: The Unity of Discovery and Invention* as an application of Meta-Theorem II in the domain of the philosophy of logic.

Pathway Two: Cognitive Science Readers

For readers interested in cognitive science, psychology, and linguistics, begin with the cognitive-scientific anchorings of Meta-Theorem II (Chapter 4, Section 5), then read the relevant sections of Piaget's *Principles of Genetic Epistemology* and Merleau-Ponty's *Phenomenology of Perception*, then enter the cognitive theorems of *Noetic Ecology* (Theorem E25 on the stratified model of consciousness, Theorem D23 on cognitive escape, Theorem F31 on the consensual manifestation of intelligence), and finally take *Generalized Evolution: The Accelerated Evolution of Human Civilization in the Noetic Ecology* as a supplement from the macro-evolutionary perspective.

Pathway Three: Civilizational Diagnostics Readers

For readers interested in social critique, civilizational evolution, and the sociology of knowledge, begin with this paper's "Explaining the Origin of the Subject-Object Illusion" (Chapter 4, Section 4, Test Four), then read *A Critique of Academic Jargon: The Alienation and Antagonism of Modern Knowledge Production* and *The Absence of Antagonism: System Dynamics and the Cognitive Roots of Civilizational Imbalance*, then enter the phase-transition models and civilizational state functions of *Civilization Dynamics*, and finally take *The War of Operating Systems: The Paradigm Conquest of Analytic Philosophy by System Ontology and the Rebirth of Philosophy* as a declaration of paradigm revolution.

V. Concerning the Author's Theoretical System

This paper is a meta-theoretical work within the author's "Noetic Trilogy." The structure of the trilogy is as follows:

The Foundational Layer (Meta-Theoretical Layer): *System Ontology: The Ultimate Jurisprudence of Existence*. Reveals the ultimate conditions for the existence of any system whatsoever—the eternal coupling of self-organizing tendency and systemic constraint. Contains two meta-axioms, six system axioms, and seven core theorems. Meta-Theorem II, discovered in this paper, stands alongside Meta-Theorem I (The Eternal Dynamical Cycle of Existence) as one of the two integrative statements of the meta-theoretical layer.

The Intermediate Layer (General Theory Layer): *Noetic Ecology: A Systems Theory of Cognition, Culture, and Society*. Applies System Ontology to the domains of cognition, culture, and society, deducing thirty-six core theorems covering the model of consciousness, cognitive escape, the transmission of noetic genes, the identification of system parasites, and synergistic cognitive networks. Provides the core concepts and qualitative theorems for *Civilization Dynamics*.

The Top Layer (Applied Theory Layer): *Civilization Dynamics: Planetary Civilization Science*. Applies Noetic Ecology to the scale of civilization, establishing civilizational state functions, phase-transition models, and computable dynamical equations. Diagnoses civilizational health and designs

blueprints for civilizational architecture.

The three books share a single set of core terms (self-organizing tendency, systemic constraint, matter/noetic duality, cognitive framework, noetic gene, system parasite, etc.), with clear logical dependencies: *System Ontology* is the "constitution," *Noetic Ecology* is the "departmental law," and *Civilization Dynamics* is the "judicial interpretation of concrete cases."

In addition, the author has completed the following specialized papers: *The Duality of Logic: The Unity of Discovery and Invention*; *How Philosophy Becomes Science?—A Formalized Foundational Mechanism Based on System Ontology*; *Generalized Evolution: The Accelerated Evolution of Human Civilization in the Noetic Ecology*; *The War of Operating Systems: The Paradigm Conquest of Analytic Philosophy by System Ontology and the Rebirth of Philosophy*; *A Critique of Academic Jargon: The Alienation and Antagonism of Modern Knowledge Production*; *The Absence of Antagonism: System Dynamics and the Cognitive Roots of Civilizational Imbalance*.

All of the above works and papers are freely available at the author's academic repository: <https://github.com/wenhe1994/wenhe001>

VI. Acknowledgments

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Finally, I thank you who have taken up this paper. The legitimacy of this paper lies not in the hands of the author, but in the hands of you, the external arbiter. If you discover a flaw, point it out. If you find a better answer, speak it. This is the spirit of Boundary-Driven Abductive Locking, and what philosophy, at its best, should be.

Guorui He

In the Meta-Level of Thought

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